PHYSICAL CHEMISTRY CUMULATIVE EXAM

February 7, 1998

Draw a schematic representation of the bound energy levels of the H atom.

A. For the lower levels (1s, 2s, 2p, 3s, 3p, 3d) discuss the following features:

(a) the energies (relative to infinite separation of the proton and the electron)

(b) the degree of degeneracy

(c) the angular momentum

(d) the average distance from the nucleus (qualitatively)

B. What happens to the energy if the main quantum number increases to very high values?

C. How do the following qualities change if the nuclear charge Z is increased (e.g. doubled in He^+, tripled in Li^{2+},...). Be quantitative if possible, i.e. specify if the quantity in question is proportional to Z, or to Z^2, or $\frac{1}{Z}$, etc.

(a) energy

(b) angular momentum

(c) average distance from the nucleus

D. Prepare a picture of the electron density along a line originating at the nucleus and extending to large distances. Include a length scale in Å.